THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A student finds that a colourless liquid turns blue litmus red and does not affect red litmus. It also fizzes with baking soda. The liquid is: 1

(a) neutral **(b)** a base **(c)** a salt **(d)** an acid

2. A bulb rated 6 V is connected to a battery of three 1.5 V cells in series. The bulb will: 1

(a) fuse **(b)** glow normally **(c)** not glow at all **(d)** glow dimly

3. A shiny solid X conducts electricity, is hard, and its oxide turns red litmus blue. X is probably: 1

(a) magnesium **(b)** carbon **(c)** sulphur **(d)** phosphorus

4. A student heated a blue crystalline solid. It turned white and drops of liquid appeared on the cooler part of the tube. On adding water to the white powder, it turned blue and became warm. This indicates: 1

(a) burning **(b)** an irreversible change

(c) a physical change **(d)** a reversible chemical change

5. A 14-year-old girl is shorter than her 14-year-old male cousin. The best explanation is: 1

(a) she is old

(b) growth rates vary and the cousin may have started his spurt

(c) she is ill

(d) she is not eating

6. Two rods A and B of equal size are heated at one end. A's pins fall in 20 s, B's pins fall in 60 s. This shows that: 1

(a) A is a better conductor **(b)** B is a better conductor

(c) both equal **(d)** neither conducts

7. A train 150 m long crosses a platform 250 m long in 20 s. The speed of the train is: 1

(a) 7.5 m/s **(b)** 12.5 m/s **(c)** 20 m/s **(d)** 8 m/s

8. A person has a blocked bile duct. The food whose digestion will be most affected is: 1

(a) water **(b)** fat-rich food

(c) starch-rich food **(d)** protein-rich food

9. A leaf is kept in the dark for 48 hours, then half of it is covered with black paper and kept in sunlight for 6 hours. Which statement is correct? 1

(a) the covered half turns blue-black
(b) the covered half remains brown with iodine
(c) both turn blue-black
(d) both remain brown

10. A ray of light falls on a plane mirror along the normal. The ray: 1

(a) passes through the mirror **(b)** travels along the mirror surface
(c) returns along the same path **(d)** is absorbed

11. If the Earth's axis were tilted by 45° instead of 23.5°, the seasons would be: 1

(a) milder **(b)** the same **(c)** more extreme **(d)** absent

12. In an experiment, 20 mL of a base is neutralised by 20 mL of a certain acid of the same type. If 40 mL of the same base is to be neutralised, the acid needed is: 1

(a) 80 mL **(b)** 10 mL **(c)** 20 mL **(d)** 40 mL

13. A student claims: 'Pure water conducts electricity well'. How would you test and correct this? 1

(a) heat it
(b) test with a tester; add salt to water to make it conduct
(c) water is a good conductor
(d) add sugar to make it conduct

14. Which of these metals will give a vigorous reaction with dilute HCl? 1

(a) silver **(b)** copper **(c)** gold **(d)** magnesium

15. Which of these is the most reliable test to decide whether a change is chemical? 1

(a) change in size **(b)** change in shape
(c) change in position **(d)** formation of a new substance

16. A boy's voice is 'cracking'. This is temporary because: 1

(a) he is ill **(b)** larynx is growing and adjusting
(c) he is cold **(d)** voice is lost

17. **Assertion (A):** Turmeric paper turns red when soap solution is dropped on it. 1
Reason (R): Soap solution is basic in nature.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** A bulb glows when the switch in its circuit is closed. 1
Reason (R): A closed switch completes the circuit so that current can flow.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** Aluminium can be beaten into very thin foils. 1
Reason (R): Metals are malleable.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** Melting of wax is a physical change. 1
Reason (R): No new substance is formed when wax melts.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. When a few drops of lemon are added to a basic solution containing turmeric (red), the colour becomes yellow. Explain. 2

22. Explain why the filament of a bulb is made of a thin tungsten wire and not of copper. 2

23. Predict the products when zinc reacts with dilute sulphuric acid and write the equation. 2

24. A student weighs an iron nail before and after rusting and finds an increase in mass. Explain. 2

25. Explain why the endocrine system is called the 'chemical messenger system'. 2

26. Explain why heating a liquid from the top does not heat it quickly but from the bottom does. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Student X finds that 10 mL of acid is neutralised by 20 mL of base. How much acid neutralises 60 mL of base of the same strength? 3

28. A student connects 6 cells of 1.5 V each in series. Find the total voltage and say what happens to a 3 V bulb connected to it. 3

29. A student has an unknown sample. It is shiny, ductile, burns in air to a white powder that turns red litmus blue. Identify the class and give a probable example. 3

30. A student kept three iron nails: A in a closed jar with dry air and calcium chloride, B in boiled water with oil layer, C in a open beaker with tap water. Predict and justify results. 3

31. Explain the effect of removing the pituitary gland in a child on growth and on other glands. 3

OR

In an experiment, 100 mL of hot water at 80 °C is poured into three cups: metal, ceramic and thermocol. After 10 min the temperatures are 55 °C, 62 °C and 72 °C respectively. Arrange them and explain. 3

32. A train 120 m long crosses a pole in 12 s and a platform in 30 s. Find its speed and the length of the platform. 3

OR

Explain the effects on the digestive system if the liver, pancreas and gall bladder are removed. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** After eating too many spicy samosas at a mela in Lucknow, Rohan complains of burning in his stomach. His mother gives him a spoonful of milk of magnesia. Rohan's younger sister suggests giving him lemon juice instead, saying it is 'good for the stomach'. 5
(i) Name the acid naturally present in the stomach.
(ii) What is the nature of milk of magnesia?
(iii) Name the type of reaction that gives relief to Rohan.
(iv) Explain why lemon juice would not be a good choice.

34. **Case study.** Ananya's torch has two cells of 1.5 V each placed one after another, a small bulb and a push switch. One day, even after pressing the switch, the bulb does not glow, though the cells are new. 5
(i) What is the total voltage of the two cells joined in series?
(ii) Name the component that opens or closes the circuit.
(iii) Name the thin wire inside the bulb that glows.
(iv) Give one possible reason for the bulb not glowing and explain it.

35. You are given samples of sulphur, iron, carbon (graphite) and aluminium. Plan tests to identify each using properties and chemical reactions. 5

36. Design an experiment to find out how three factors (air, water and salt) affect rusting, using six nails; describe the set-up, observations and conclusions. 5

OR

A school health survey finds that many students skip breakfast, sleep late and use mobiles at night. Analyse the consequences and prepare an action plan. 5

37. Design a project to compare the heat insulation of three materials (wool, aluminium foil, paper) using identical beakers of hot water. Describe the variables, procedure and expected result. 5

OR

A student plots the following data: time (s): 0, 5, 10, 15, 20; distance (m): 0, 10, 20, 20, 50. Describe the stages of motion, find the speed in each stage and the average speed. 5

38. Design an investigation to find how the breathing rate changes with activity and present the data in a table with a conclusion. 5

OR

Design a fair test to find the effect of light intensity on the rate of photosynthesis using hydrilla and a lamp. Name the variables, steps and expected graph. 5

END OF QUESTION PAPER Total: 80 Marks